Positive Square Energy for a Two-C_5 Bouquet with Arbitrary Trees

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Abstract

Let G be a connected cactus whose only cyclic blocks are two copies of C_5 having exactly one common vertex. Arbitrary trees may be attached at every vertex of this nine-vertex core. We prove the explicit strict bound

$$s^+(G) \geq |V(G)| + 1 - \frac{4}{3\sqrt{13}} > |V(G)|.$$

The proof eliminates the attached trees by exact matching belief propagation, uses a coefficientwise integer-polynomial certificate on the resulting weighted core, and integrates the phase of the normalized characteristic polynomial. The polynomial certificate is reproducible and is recorded by a canonical SHA-256 digest.

1 Introduction

All graphs are finite, simple, and undirected. If the adjacency eigenvalues of G are $\lambda_1, \ldots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \qquad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2.$$

Liu, Tang, and Zhang proved the positive and negative square-energy conjecture for connected graphs [2]. Akbari, Kumar, Mohar, Pragada, and Zhang proposed the stronger assertion $s^+(G) \geq n$ when a connected n-vertex graph has at least $n+1$ edges [1]. Optimizing the Liu–Tang–Zhang doubly nonnegative certificate block by block settles most bicyclic cacti but leaves, in particular, the pair of cyclic block lengths $\{5, 5\}$. The argument here resolves the shared-vertex member of that family.

Theorem 1.1. *Let G be a connected cactus whose only cyclic blocks are two copies of C_5, and suppose that these cycles share exactly one vertex v_0. Every edge outside the cycles may belong to an arbitrary tree attached at any vertex of the two-cycle core. If $n = |V(G)|$, then*

$$s^+(G) \geq n + 1 - \frac{4}{3\sqrt{13}} > n. \tag{1}$$

The hypothesis means a one-vertex bouquet: the two C_5 blocks intersect in the single vertex v_0. It does not cover vertex-disjoint cycles joined by a bridge or a path, and it does not assert a result for arbitrary bicyclic cacti.

The coefficient inequality used below is an equality when there are no nontrivial tree messages. The later analytic inequality $\arctan u \leq u$ is strict for $u > 0$, however, so we do not claim that the constant in (1) is the best possible spectral lower bound.

2 Signless matchings and grouped Sachs expansion

For a forest F, define its signless matching partition by

$$Z_F(t) = \sum_M t^{|V(F)|-2|M|}, \tag{2}$$

where the sum runs over all matchings of F. More generally, for positive vertex activities $a = (a_v)_{v\in V(H)}$, write

$$Z_H(a) = \sum_M \prod_{v \text{ unmatched by } M} a_v. \tag{3}$$

All edge activities are one. Thus (2) is the specialization of (3) in which every vertex activity is t.

Let $\phi_G(z) = \det(zI - A(G))$ and normalize it on the imaginary axis by

$$\Psi_G(t) = i^{-n}\phi_G(it) = \prod_{j=1}^{n}(t + i\lambda_j), \qquad t > 0. \tag{4}$$

Grouping the Sachs expansion by its collection $\mathcal{C}$ of vertex-disjoint cycle components gives

$$\Psi_G(t) = \sum_{\mathcal{C}} \prod_{C\in\mathcal{C}} \left(-2i^{-|C|}\right) Z_{G-V(\mathcal{C})}(t). \tag{5}$$

The empty collection contributes $Z_G(t)$. A 5-cycle contributes the multiplier $-2i^{-5} = 2i$. Since the two cycles under consideration share v_0, they are not vertex-disjoint and cannot occur together in a Sachs subgraph. In particular, there is no double-cycle term in (5).

3 Exact elimination of attached trees

Orient every tree branch toward its core vertex. For an oriented noncore edge $u \to p$, let $T_{u\to p}$ be the component below u, including u, and define

$$q_{u\to p}(t) = \frac{Z_{T_{u\to p}}(t)}{Z_{T_{u\to p}-u}(t)}. \tag{6}$$

The denominator is a product of child-subtree partitions and is positive for $t > 0$.

Lemma 3.1 (Rooted-tree matching recursion)**.** *For every oriented branch edge,*

$$q_{u\to p}(t) = t + \sum_{w \text{ child of } u} \frac{1}{q_{w\to u}(t)} \geq t > 0. \tag{7}$$

After all noncore vertices are eliminated, the effective activity at a core vertex v is

$$a_v(t) = t + \sum_{u \text{ attached at } v} \frac{1}{q_{u\to v}(t)} = t + y_v(t), \qquad y_v(t) \geq 0. \tag{8}$$

Moreover, all terms of the grouped Sachs expansion have the same positive prefactor

$$K(t) = \prod_{u \text{ adjacent to the core}} Z_{T_{u\to v}}(t) > 0. \tag{9}$$

Proof. Split a matching of $T_{u\to p}$ according to whether u is unmatched or is matched to one of its children. After the child-subtree partitions are factored out, this gives (7). A leaf has message t, so induction also proves $q_{u\to p} \ge t > 0$.

For a branch $T = T_{u\to v}$ rooted next to a core vertex v, a matching not using uv contributes Z_T. A matching using uv contributes Z_{T-u}. Extracting Z_T changes the latter choice into the weight

$$\frac{Z_{T-u}}{Z_T} = \frac{1}{q_{u\to v}}.$$

The incident branch choices are mutually exclusive, so their sum adds exactly the terms in (8). Independence among branches yields the factor (9).

It remains important to check cycle deletion, rather than silently canceling an unjustified ratio. If a Sachs cycle deletes a core vertex v, every branch formerly attached at v becomes a disconnected full tree and contributes Z_T, exactly its factor in $K(t)$. At each retained core vertex, the branch is absorbed into the same effective activity as before. Thus deleting all five vertices of either cycle preserves the full common factor $K(t)$. No quotient of tree partitions remains. $\square$

The same argument covers a rooted tree described as being identified with a core vertex: remove that core root and treat each resulting component as one of the branches above.

4 The weighted two-C_5 core

Label the common vertex 0. On each lobe $j \in \{1, 2\}$, label the other vertices consecutively $1, 2, 3, 4$, with activities x_1, x_2, x_3, x_4 when one lobe is under discussion. Deleting vertex 0 leaves a path P_4. Its weighted matching partition is

$$A(x_1, x_2, x_3, x_4) = x_1x_2x_3x_4 + x_3x_4 + x_1x_4 + x_1x_2 + 1. \tag{10}$$

These are respectively the empty matching, the three one-edge matchings, and the unique two-edge matching of P_4.

If vertex 0 is matched along this lobe, the sum over the two possible incident cycle edges is

$$B(x_1, x_2, x_3, x_4) = x_2x_3x_4 + x_2 + x_4 + x_1x_2x_3 + x_1 + x_3. \tag{11}$$

Indeed, either edge from 0 leaves a P_3. The two displayed groups in (11) correspond to the two distinct choices of incident edge, so no additional factor of two is present.

Let A_j, B_j denote (10)–(11) on lobe j, and let a_0 be the activity at the common vertex. Splitting core matchings according to the status of vertex 0 gives

$$R = a_0A_1A_2 + B_1A_2 + A_1B_2. \tag{12}$$

The first term leaves 0 unmatched; the other terms match it into exactly one of the two lobes. These cases are disjoint and exhaustive.

Deleting one whole C_5 from the core leaves the P_4 in the other lobe. Lemma 3.1, the $2i$ cycle multiplier, and the absence of a double-cycle term therefore give the exact identity

$$\boxed{\Psi_G(t) = K(t)\big[R + 2i(A_1 + A_2)\big].} \tag{13}$$

All quantities in brackets are positive for $t > 0$. Since $K(t)$ is positive and real, it does not affect the continuous argument. Consequently

$$\Theta_G(t) := \arg \Psi_G(t) = \arctan \frac{2(A_1 + A_2)}{R}, \qquad 0 < \Theta_G(t) < \frac{\pi}{2}. \tag{14}$$

5 The coefficientwise certificate

Put

$$D(t) = t^4 + 7t^2 + 9. \tag{15}$$

The algebraic heart of the proof is the following finite exact statement.

Proposition 5.1. *Whenever all nine core activities are at least $t > 0$,*

$$2R \geq tD(t)(A_1 + A_2). \tag{16}$$

Equality holds in (16) *when every activity is t.*

Proof. Write

$$a_0 = t + y_0, \qquad x_{j,k} = t + y_{j,k} \quad (j = 1, 2,\ k = 1, 2, 3, 4),$$

where every y is nonnegative. Substitute these expressions into (10)–(12) and form

$$P = 2(a_0A_1A_2 + B_1A_2 + A_1B_2) - t(t^4 + 7t^2 + 9)(A_1 + A_2). \tag{17}$$

Appendix A gives an exact, exhaustive coefficient verification over $\mathbb{Z}$. The expansion has 1290 nonzero monomials; all coefficients are positive integers between 1 and 22, and every monomial contains at least one y variable. Representative terms include

$$2t^8y_0, \qquad t^8y_{1,1}, \qquad 2y_{2,1}, \qquad 2y_{2,2}y_{2,3}y_{2,4}.$$

Thus $P \geq 0$ on the nonnegative orthant, proving (16). Since the part independent of all y variables is identically zero, equality holds when all y vanish. □

Combining (14) and (16), and using monotonicity of the arctangent, yields

$$0 < \Theta_G(t) \leq \arctan \frac{4}{tD(t)}. \tag{18}$$

The factor 4 is forced: the imaginary part in (13) is $2(A_1 + A_2)$, while (16) bounds $(A_1 + A_2)/R$ by $2/(tD(t))$.

6 Coulson integral and square energy

For the continuous argument fixed in (4), the signed square-energy Coulson identity is

$$s^+(G) - s^-(G) = -\frac{4}{\pi} \int_0^\infty t\Theta_G(t)\, dt. \tag{19}$$

For completeness, it follows by summing the scalar identity for the factors $t+i\lambda_j$ and integrating; the trace relation $\sum_j \lambda_j = 0$ cancels the nonintegrable first-order term at infinity. The first-quadrant location in (14) fixes the branch and precludes winding.

By (18) and $\arctan u \leq u$ for $u \geq 0$,

$$\int_0^\infty t\Theta_G(t)\, dt \leq \int_0^\infty \frac{4}{t^4 + 7t^2 + 9}\, dt. \tag{20}$$

Set $t = \sqrt{3}u$. The reciprocal-quartic formula

$$\int_0^\infty \frac{du}{u^4 + bu^2 + 1} = \frac{\pi}{2\sqrt{2+b}} \qquad (b > -2)$$

then gives the exact value

$$\int_0^\infty \frac{4}{t^4 + 7t^2 + 9}\, dt = \frac{2\pi}{3\sqrt{13}} < \frac{\pi}{2}. \tag{21}$$

Equations (19)–(21) imply

$$s^+(G) - s^-(G) \geq -\frac{8}{3\sqrt{13}}. \tag{22}$$

The graph has exactly two independent cycles, hence $|E(G)| = n + 1$. Therefore

$$s^+(G) + s^-(G) = \operatorname{tr} A(G)^2 = 2|E(G)| = 2n + 2. \tag{23}$$

Averaging (22) and (23) proves

$$s^+(G) \geq n + 1 - \frac{4}{3\sqrt{13}} > n,$$

because $4 < 3\sqrt{13}$. This proves Theorem 1.1.

Remark 6.1 (Equality nuance). The coefficientwise inequality (16) is exact at the bare core, where every activity equals t. Nevertheless, $\arctan u < u$ for every $u > 0$, and the phase in (14) is positive. Thus the displayed spectral bound is not an equality even for the bare core, and no optimality claim for its constant is made.

A Exact certificate and reproducibility

The certificate script is

`positive-square-energy/experiments/c5_bouquet_matching_certificate.py`.

Run it from the repository root with

```
python positive-square-energy/experiments/c5_bouquet_matching_certificate.py
```

It uses SymPy to construct (17) as a polynomial over $\mathbb{Z}$ in the ordered variables

$$(t, y_0, y_{1,1}, y_{1,2}, y_{1,3}, y_{1,4}, y_{2,1}, y_{2,2}, y_{2,3}, y_{2,4}).$$

It then asserts all of the following:

1. the expanded polynomial has exactly 1290 nonzero terms;
2. every coefficient is nonnegative (in fact the minimum is 1 and the maximum is 22);
3. substitution of zero for all nine y variables gives zero;
4. every exponent vector has a nonzero y coordinate; and
5. the SHA-256 digest of the canonical stream `monomial:coefficient`, one term per line in SymPy's ordered term order, is

4c436cac772395d2a8edfdd81408ffe426759d3e94d66df2e4ab0235a3343110.

The script prints

```
PASS two-C5 bouquet matching coefficient certificate
terms=1290 min_coefficient=1 max_coefficient=22
y_constant=0 all_coefficients_nonnegative=True
sha256=4c436cac772395d2a8edfdd81408ffe426759d3e94d66df2e4ab0235a3343110
```

This is an exhaustive symbolic verification, not a numerical sampling of the nonnegative orthant. Since coefficientwise nonnegativity is itself the proof of Proposition 5.1, the script is part of the exact proof. For file-level provenance, the script file used for this manuscript has SHA-256 digest

53e5ede12064953fe3d19aa173247e63ea3ab17d95ca2901d8f5707dae2e9f92.

References

[1] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.

[2] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.